Non-Abelian Symmetries

Introducing Non-Abelian Symmetries

Consider the theory of φ_1 and φ_2 (real, scalar) specified by:

$$\mathcal{L} = -\frac{1}{2}\partial^\mu \varphi_1 \partial_\mu \varphi_1 - \frac{1}{2}\partial^\mu \varphi_2 \partial_\mu \varphi_2 - \frac{1}{2}m^2(\varphi_1^2 + \varphi_2^2) - \frac{1}{16}\lambda(\varphi_1^2 + \varphi_2^2)^2$$

We can generalize this for N real scalar fields φ_i with:

$$\mathcal{L} = -\frac{1}{2}\partial^\mu \varphi_i \partial_\mu \varphi_i - \frac{1}{2}m^2 \varphi_i \varphi_i - \frac{1}{16}\lambda(\varphi_i \varphi_i)^2$$ $\longrightarrow$ Repeated indices are summed.

This lagrangian is invariant under the SO(N) transformation: $\varphi_i(x) \longrightarrow R_{ij}\varphi_j(x)$

Verification:

$$\mathcal{L} = -\frac{1}{2}\partial^\mu (R_{ij}\varphi_j)\partial_\mu (R_{ij}^T \varphi_j) - \frac{1}{2}m^2(R_{ij}\varphi_j)^2 - \frac{1}{16}\lambda[(R_{ij}\varphi_j)^2]^2$$ $\longrightarrow$ $(R_{ij}\varphi_j)^2 = \varphi_i^2$ since rotation cannot change the magnitude.

$$= -\frac{1}{2}\partial^\mu \varphi_i \partial_\mu \varphi_j - \frac{1}{2}m^2(\varphi_j \varphi_j) - \frac{1}{16}\lambda[\varphi_i \varphi_j]^2$$ $\longrightarrow$ Same as for φ_i since indices are summed.

This Lagrangian is also invariant under the Z_2 transformation: $\varphi_i(x) \longrightarrow -\varphi_i(x)$, This transformation extends the group from SO(N) to O(N). However, for the time being we will only focus on the SO(N) symmetry.

To proceed further, we need to know a bit of group theory.

Consider an infinitesimal SO(N) transformation: $R_{ij} = \delta_{ij} + \Theta_{ij} + O(\Theta^2)$ $\nearrow$ Anti-symmetric due to the ortogonality of R_{ij}.

We can express Θ_{ij} in terms of a basis set of Hermitian matrices $(T^a)_{ij}$

Here a runs from 1 to $\frac{1}{2}N(N-1)$ which is the number of linearly independent, anti-symmetric $N \times N$ matrices.

Lets make some sense of this,

First off we want basis matrices which by definition implies linear independence. We also want these matrices to be anti-symmetric as Θ_{ij} is anti-symmetric.

$N \times N$ obviously because the SO(N) group requires mat.

These matrices obey the normalization condition:

$$Tr(T^a T^b) = 2\delta^{ab}$$

And we can write: $\Theta_{jk} = -i\Theta^a (T^a)_{jk}$ $\nearrow$ we are basically expressing Θ_{jk} in another basis, one described by the basis vectors T^a.

$\downarrow$ set of $\frac{1}{2}N(N-1)$ real (cuz Θ_{jk} is real), infinitesimal parameters.

We call T's the generator matrices of $SO(M)$.

The product of any two $SO(M)$ transformations is another $SO(M)$ transformation which implies:

$[T^a, T^b] = i f^{abc} T^c$ → This equation specifies the Lie Algebra of the group.

↓

Structure coefficients of the group.

The group is abelian if $f^{abc} = 0$ otherwise it is non abelian.

$SO(M)$ for $M \geq 3$ is non-abelian.

Now, if we multiply the commutator above by T^d then take the trace while imposing: $Tr(T^a T^b) = 2\delta^{ab}$, we get:

$$f^{abd} = -\frac{1}{2} i \, Tr([T^a, T^b] T^d)$$

↳ Can be proven to be real and anti-symmetric.

In the case of $SO(3)$ where the generators are given by: $(T^a)_{ij} = -i\epsilon^{aij}$ → Completely antisymm. Levi-Civita.

With $\epsilon^{123} = +1$, we get the commutation relations as:

$$[T^a, T^b] = i\epsilon^{abc} T^c$$

Therefore, $f^{abc} = \epsilon^{abc}$

The SU(M) Group

Now, consider a theory with M complex scalar fields φ_i:

$$\mathcal{L} = -\partial^\mu \varphi_i^\dagger \partial_\mu \varphi_i - m^2 \varphi_i^\dagger \varphi_i - \frac{1}{4}\lambda(\varphi_i^\dagger \varphi_i)^2$$

This lagrangian is invariant under the $U(M)$ transformation: $\varphi_i(x) \to U_{ij}\varphi_j(x)$ ↗ Unitary Matrix

We can write: $U_{ij} = e^{-i\theta}\tilde{U}_{ij}$ ↗ Unitary matrix with det +1

↳ Special unitary matrix

Here, since U_{ij} is unitary, we have basically just expressed it essentially in polar form.

$e^{-i\theta}$ belongs to $U(1)$ and $\tilde{U}_{ij}$ belongs to $SU(M)$. Thus we can say:

$$U(M) = U(1) \times SU(M)$$

Now, consider an infinitesimal $SU(M)$ transformation:

$$\tilde{U}_{ij} = \delta_{ij} - i\omega^a (T^a)_{ij} + O(\omega^2)$$

T^a is hermitian and traceless due to the properties of $\tilde{U}_{ij}$. The index runs from 1 to M^2-1 which is the number of linearly independent, hermitian, traceless, $M \times M$ matrices.

We can choose matrices that obey: $Tr(T^a T^b) = 2\delta_{ab}$

Its interesting to note that the generators for $SU(2)$ (the Pauli matrices) have structure coefficients of

$2\epsilon_{abc}$ which is of the same form as $SO(3)$.

Now, how do we choose the generators?

First we account for the $SO(N)$ generators $\left(N(N-1)/2\right)$ these have one $+i$ and one $-i$ above and below the main diagonal. We can get another set by putting $+1$ above and below the main diagonal. There are $\frac{N}{2}(N-1)$ of these as well. There are also diagonal matrices with n 1's along the diagonal followed by a $-n$ followed by 0's. We will need a normalization constant to satisfy the trace condition but that is irrelevant. There are $N-1$ such generators making the total N^2-1.

Let's loop back and look at the Lagrangian. The thing is that we can express the complex fields as φ_1 and φ_2. Thus, the theory obeys a larger symmetry of $SO(2N)$.